

MANUEL FRANA

Senior QA Automation Engineer | SDET

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SUMMARY

Senior QA Automation Engineer with 5+ years building and testing the core backend framework services used by every product team at an enterprise SaaS company. Work spans the full software development life cycle (SDLC): I identify quality gaps, author the technical specification, and implement the fix in production code myself — operating as a hybrid SDET / software engineer. Because the platform runs on a proprietary in-house framework, I build the test harnesses, instrumentation and release tooling that off-the-shelf frameworks would otherwise provide. Strong in test automation, API and integration testing, CI/CD quality gates, progressive feature rollouts and root cause analysis across PHP, Python and JavaScript services backed by MySQL and Redis.

EXPERIENCE

Senior Quality Assurance Engineer — Framework Team

Oct 2022 – Present

Avature

Remote, Spain

- Own quality for the **core backend framework** — database access layer, **Redis** caching, and metrics registration — consumed by **40** product engineering teams, combining test strategy ownership with hands-on feature development in **PHP**, **Python** and **JavaScript**.
- Independently identify coverage gaps and architectural risks, author the technical specification and acceptance criteria, then **implement and ship the change end to end** in production code — removing the handoff between QA analysis and development.
- Re-architected monolithic code-style checks that scanned the entire codebase into **module-scoped, independently schedulable test units**, unlocking **parallel execution** and cutting suite runtime by up to **50%**.
- Design and maintain **automated test suites** spanning **unit, integration, API and end-to-end (E2E)** layers with **Selenium WebDriver** and **custom in-house test harnesses**, wired into **CI/CD pipelines** on **Docker** and **Kubernetes** so **quality gates** block regressions before merge.
- Own the **progressive rollout plan** for a new framework feature — enabling it incrementally across client instances and **measuring performance and production impact** at each stage before wider release.
- Automate operational procedures across server environments with **Rundeck** jobs, giving QA and development **self-service execution** of tasks that previously required manual, error-prone intervention.
- Extend the framework's **metrics instrumentation and observability** layer, surfacing dashboards that detect **performance regressions** in caching and database access paths before they reach production.
- Coordinate a **weekly working session with 2 QA engineers** to review in-flight activity, unblock technical questions and **prioritize testing and automation work** across the team's backlog, in an **Agile/Scrum** environment.

Quality Assurance Engineer

Jun 2021 – Oct 2022

Avature

Remote, Argentina

- Built **automated regression and E2E test suites** with **Selenium** and unit tests, replacing **20** recurring manual regression passes per release cycle.
- Executed **exploratory, data consistency and regression testing** across releases and authored reproducible **defect reports**, driving them to resolution with development teams.
- Validated backend behaviour by inspecting **HTTP request/response traffic** and querying **MySQL** directly to confirm **data integrity** across services.

PROJECTS

Catox — Multiplayer Strategy Game | *Luau, React (jsdothua), Rodux, Rojo, Wally, TestEZ*

2026 – Present

- Architected a **server-authoritative multiplayer game** (~**49K LOC** Luau, live on Roblox) in which every state transition is validated server-side, eliminating client-side exploits.
- Implemented a **Command pattern gateway** routing all RemoteEvents and RemoteFunctions through a single **validation, authorization and abuse-detection** layer, centralizing security checks instead of scattering them across call sites.

- Maintain a **67-spec automated test suite** (TestEZ + Lune) executed headlessly, including an **architectural regression test** that fails the build if any code bypasses the event gateway.
- Built a component-based UI with **React** and **Rodux** state management, opponent AI with pluggable scoring strategies benchmarked through an **A/B experiment harness**, and funnel analytics and match telemetry to measure onboarding drop-off.

Courts — Booking Automation Bot | *Python, Playwright, asyncio, SQLite, Docker* 2026

- Built an **asynchronous Telegram bot** that scrapes live tennis court availability and **auto-reserves** slots the moment they are released at midnight, automating a manual booking race for a group of players.
- Reverse-engineered an ASP.NET booking platform's internal APIs (ViewState / EventValidation, session-bound tokens) and drove authenticated flows with **Playwright** and **httpx**; encrypted credentials at rest with **Fernet (AES)** and persisted sessions, reservations and cached availability in **SQLite**.
- Diagnosed a production outage through **root cause analysis** — booking tokens fetched in one HTTP session were rejected when submitted from another — shipped the same-session fix and documented it in a written **post-mortem**.
- **Containerized with Docker** and deployed with a persistent volume and scheduled job queue; covered availability parsing with **pytest** against recorded API fixtures.

TECHNICAL SKILLS

Languages: PHP, Python, JavaScript/TypeScript, Luau, Java, SQL, HTML/CSS, Bash
Testing & QA: Test Automation, Test Automation Frameworks, Selenium WebDriver, Playwright, pytest, Unit Testing, Integration Testing, API Testing, End-to-End (E2E) Testing, Regression Testing, Exploratory Testing, Parallel Test Execution, Test Plans & Test Cases, Defect Tracking, Root Cause Analysis, Shift-Left Testing, Postman, Charles Proxy
Backend & Data: REST APIs, MySQL, Redis, Caching Strategies, SQLite, Database Design, Data Consistency Validation
DevOps & Observability: Docker, Kubernetes, CI/CD Pipelines, Rundeck, Git & GitHub, Linux, Metrics Instrumentation, Log Analysis, Monitoring & Alerting
Practices: Agile / Scrum, SDLC & STLC, Release Management, Progressive / Phased Rollouts, Code Review, Technical Specification Writing, Cross-Functional Collaboration, Mentoring

EDUCATION

Universidad Tecnológica Nacional (UTN) — Córdoba Graduated Jun 2024
Engineer's Degree, Information Systems Engineering (5-year program) — GPA 8.67 / 10 Córdoba, Argentina

- Successfully cleared all terminal exams on the first attempt.

Cambridge University — First Certificate in English (FCE)
England

LANGUAGES

Spanish — Native | English — Advanced (professional working proficiency)